

‘Who Likes What and, Why?’ Insights into Modeling Users’ Personality Based on Image ‘Likes’

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Abstract—The increased proliferation of data production technologies (e.g., cameras) and consumption avenues (e.g., social media) has led to images and videos being utilized by users to convey innate preferences and tastes. This has opened up the possibility of using multimedia as a source for user-modeling. This work attempts to model personality traits (based on the Five Factor Theory) of users using a collection of images they tag as ‘favorite’ (or like) on Flickr. First, a set of semantic features are proposed to be used for representing different concepts in images which influence users to like them. The addition of the proposed features led to improvement over state-of-the-art by 12 percent. Second, a novel machine learning approach is developed to model users’ personality based on the image features (resulting in upto 15 percent improvement). Third, efficacy of the semantic features and the modeling approach is shown in recommending images based on personality modeling. Using the modeling approach, recommendations are made regarding the factors that might influence users with different personality traits to like an image.

Index Terms—Big five factor, personality modeling, semantic features, ordinal regression, image recommendation

1 INTRODUCTION

THIS work addresses the problem of computationally modeling users’ personality based on images they ‘like’. With the advent of social media, multimedia has become an interaction channel among users. Users convey innate preferences and tastes through implicit feedback (‘likes’ on Facebook, ‘favorite’ on Flickr etc.), which leave adequate cues to predict users’ personality [36].

Images contain several concepts, which can be represented by features such as objects, scenes, faces, colorfulness etc. Users’ ‘likes’ on images can be attributed to the presence/absence of these features (and their combinations). For example, one user may prominently like colorful images with a lot of people while another user may like black and white images of natural landscapes. It would be interesting to investigate if there is a relationship between such preferences (‘likes’) and users’ personality. Given a dataset of images liked by certain users, can we identify the concepts which influence the users with different personality traits to like certain images? If yes, can we computationally model

this relationship thereby automating users’ personality assessment? These are some of the questions that motivate this work.

To assess users’ personality, researchers in psychology have developed several psychometric tests. One such test is to let the users answer a questionnaire and convert the answers into a score on different traits of users’ personality (conscientiousness (C), openness to experience (O), agreeableness (A), extroversion (E) and neuroticism (N) according to the Five Factor Theory [17]). Such quantitative measures of personality traits have been found to be consistent with many user behaviors [50]. However, the downside of such psychometric tests is that users have to explicitly give their feedback to questionnaires. By automating this process of personality assessment, several domains can be enhanced—personalizing recommender systems, increasing acceptance of HCI systems, adapting assistive technologies involving robots, and so on. These applications can be enabled by taking into consideration users’ innate qualities that explain *why* they behave in certain ways.

There are two dimensions to automatic personality prediction [74]: a) *personality recognition* and b) *personality perception*. While automatic *personality recognition* is about inferring self-assessed personality traits, automatic *personality perception* is about inferring the personality that observers attribute to a given user. A comprehensive review of these concepts is presented in [74]. Often it is found that there is a lot of data related to users (such as videos/products liked/rated on social media and other websites), but no information about their personality profiles. It is also infeasible, in many cases, to ask users to answer the personality questionnaires. However, it is less troublesome to get assessors to build the personality profile of the users by asking them to examine users’ data. This can be a huge step forward in non-intrusive personality acquisition

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techniques, which has been shown to be preferred the most by users in [26]. In this work, we focus on modeling the self-assessed personality traits.

Most of the works in automatic personality modeling take the approach of extracting features from data (related to users) and using them directly to model personality of users (termed as Features-to-Personality—F2P). However, research in psychology [62] shows that personality is better ascertained by asking users a set of well-designed questions. The answers to these questions are then converted into scores for five personality traits. Taking this approach (F2A+A2P: Features to Answers + Answers to Personality) with features extracted from data was shown to be more effective than predicting the personality scores directly [67]. In this work, we evaluate both these approaches towards personality modeling. *Based on the images that different users like, we aim to predict their personality along these five traits.*

Section 2 describes related works in personality computing literature and the prior work in personality modeling based on implicit feedback on images. Section 3 proposes a new set of features that complement feature sets in existing literature. A novel personality modeling approach based on these features is presented in Section 4. Section 5 presents an evaluation and analysis of both feature representation and personality modeling in the context of recommender applications.

2 BACKGROUND AND RELATED WORK

2.1 Personality Computing

Brunswick Lens model [8] describes that personality is externalized through any behavior that others can perceive. Studies in psychology show that to get strong personality cues, users should be given the necessary freedom of control and motivation to express themselves through that behavior [19]. Many experimental studies have shown that photographs, being universal in expression even when users speak different languages, contain valid personality-related cues [6], [20], [21], [54], [55], [65]. Similar works were conducted on profile photos and images of users posted on social networks which talk about personality expression in images [25], [37]. Thus, with the emergence of multimedia as a new tool for self-representation and self-expression, the possibility of systematically assessing users' personality has opened up.

Researchers used cues like choice of words in language [3], intonation of voice while speaking [46], [51], type of people one befriends [16], [36], multi-modal information such as facial expressions, prosodic features and textual features from subtitles, [2], [9], [67] to computationally model users' personality (see [74] for a thorough review). These works use explicit information about the users (such as their facial expressions, speech prosody features, textual features etc., from users) to model their personality. To the best of our knowledge, works which use implicit feedback (for example, characteristics of images 'liked') to model users' personality are [11], [22]. We explain them as a part of the related work Section 2.2. However, there have been works like [36] which show that implicit feedback can predict users' personality reliably. *The factor that differentiates our work from [36] is that they just use the implicit feedback matrix by the users on several categories like movies, books etc. However we attempt*

to model users' personality based on the characteristics (features) associated with images they like.

Five Factor Model (FFM) [17] is a widely used personality model. Users' personality, according to FFM, consists of: conscientiousness (C), openness to experience (O), agreeableness (A), extroversion (E) and neuroticism (N). One of the inventories used to measure these traits is the BFI-10 [62]. There are studies [32], [39], [70] which show that BFI-10 has good correlation with other independent indices measured simultaneously with the personality test [50]. For example, it was seen that a person who had a high score on extraversion had more social contacts [78] and the correlation between different personality traits and speech traits was studied in [52], [59].

2.2 Personality Modeling Based on Image 'Likes'

Images tagged as 'favorite' (or liked) by a group of Flickr users were collected in [11]. The users were asked to answer the BFI-10 questions [62] to get their self-assessed personality profile. The process of automatically assessing the personality profile of users involved learning a regression model (LASSO [71]) mapping 'low-level' image features to their personality profile. The difference in self-assessment and others-assessment of personality was studied and the following observations were made [11]:

- Others-assessment depends on the consensus among different external assessors and therefore, with a reasonable level of accuracy, can be computationally predicted.
- Self-assessment depends on an individual assessment and the learned model does not generalize well.

Similar observations, regarding predicting self-assessed and others-assessed profiles, were also reported in other works [45], [46]. The reason for high performance at predicting others-assessed personality scores is that machines and assessors have access to the same information and can only work within the bounds of limited information provided to them, in this example the behavior of users' 'likes' on images. However, this does not apply to predicting self-assessed scores as users' assessment of their personality is based on factors which certainly go beyond the images they like (for instance their personal experiences). Therefore, if factors which are likely to better represent a users' self-assessed personality profile are considered, it might improve the prediction performance.

F2A+A2P Approach. We take an F2A+A2P approach that has been shown [67] to model personality traits better than the F2P approach. However there are some differences with [67] distinguishing the application of the F2A+A2P approach to our situation. F2A+A2P was used in [67] where the F2A stage was implemented using Sparse and Low-rank Transformation (SLoT) for predicting the personality scores of people inside movies. This was done using explicit cues (such as facial expressions, prosodic features etc.) about the subjects whose personality was being modelled from the movies. *However, we propose a novel machine learning technique Feature Selection based Ordinal Regression: FS-OR (described in Section 4) to model the F2A stage using implicit cues (from images liked by users).*

This requires investigating a diverse set of features to represent

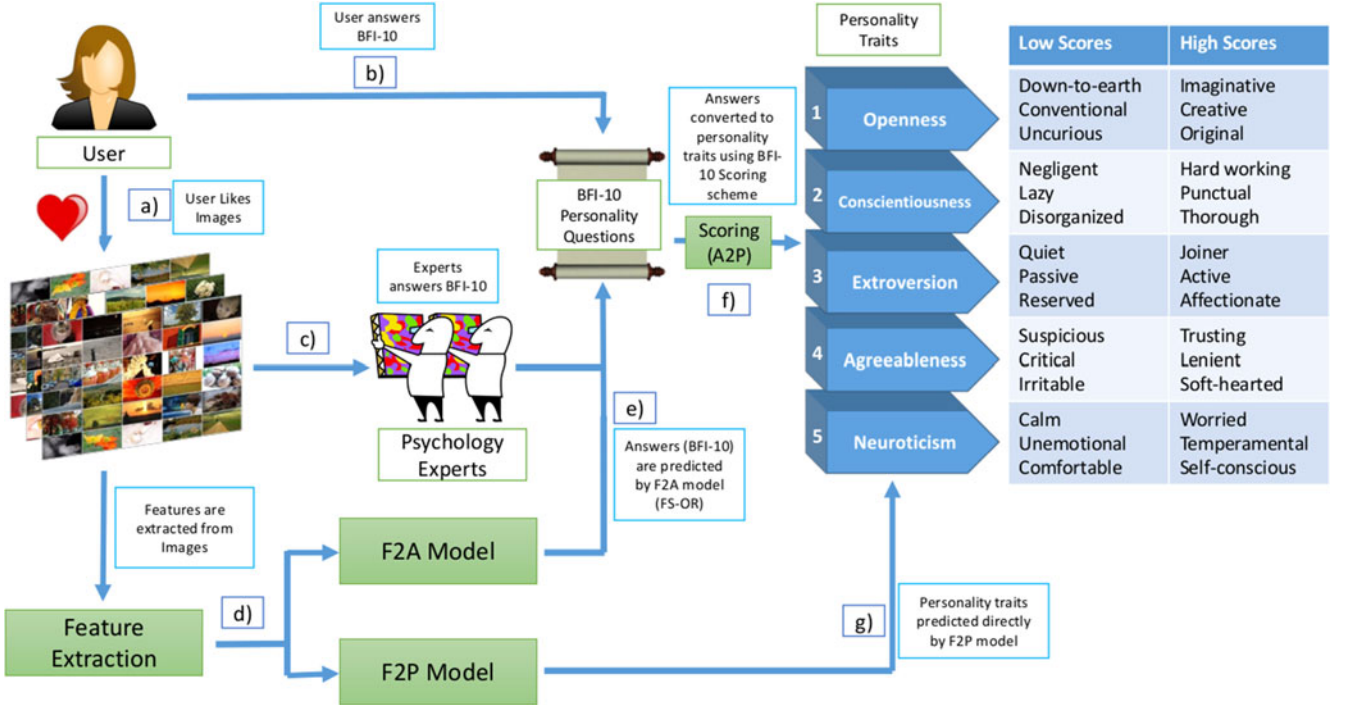


Fig. 1. **Proposed pipeline of Features to Answers and Answers to Personality** is described in steps a-f: F2P approach is used by current computational models in Personality Computing. We compare its performance with our proposed F2A+A2P (FS-OR) approach.

the users' personality expressed through their likes (which is more challenging than modeling users' personality through explicit cues such as facial expressions etc.).

Personality prediction is thus divided into two stages (shown in Fig. 1), namely transforming features to answers (F2A) and mapping the answers to trait scores (A2P). For the first stage, i.e., transforming the feature space into answers, FS-OR is used (which is an improvement over our earlier work [22]). The motivation for using this approach is two-fold. (a) Each personality trait (and answers corresponding to its calculation) might only be influenced by a few features (motivating the feature selection in the machine learning pipeline). (b) The values to answers are ordinal in nature (motivating the use of ordinal regression in formulating the problem). And for the second stage (A2P), BFI-10 scoring scheme [62] is used to predict the personality profiles.

Research Questions Investigated. Specifically, we attempt to answer the following questions:

- RQ 1. Identify image feature representations that best address the problem of modeling users' personality based on images they 'like'. (Section 3)
- RQ 2. How to model users' personality based on these image representations (features)? (Section 4)
- RQ 3. What concepts (features) are most important to model different personality traits? (Section 5.3)

The above questions are answered in the context of a recommender application that models user image association based on implicit feedback ('likes'). The recommender models user image interactions to serve in two scenarios:

- Given a user (represented by personality profile), what kind of images will he/she like?
- Given an image, find the user/group (represented by a personality profile), who will like it.

Contributions. This work investigates the problem of automatic personality prediction of users based on collections of images that have been liked by them. The specific contributions of this work are. (1) We present a new (larger) set of 'high-level features' (semantic) features and demonstrate their relevance in modeling personality profiles. (2) We adopt the F2A+A2P approach to modeling personality to work under implicit ('likes' on images) user information. We propose a novel method Feature Selection based Ordinal Regression (FS-OR) that models the F2A mapping by factoring in the ordinal relationship between the rating scale of answers. However, our main contribution is to push the state-of-the-art in predicting users personality profiles using images they like.

3 ADDING SEMANTIC FEATURES

We first describe the intuition behind using 'high-level' semantic features, in this section. The aim of these features is to provide a better representation of users' likes and how they relate to different personality profiles. We then describe the features used in our work (see Table 1 for a complete list).

3.1 Need for Semantic Features to Capture Individual Preferences

Depending on a person's psychophysical nature, one is drawn to different 'kinds' of images. To make this distinction, some amount of domain knowledge has to be included to capture the differences in what different people like. For example, the left of the collage (Fig. 2) shows images liked by a user with high score on conscientiousness. This trait reflects in socially prescribed impulse control and a sense of thoughtful behavior. Images liked by such users consist of carefully planned and timed shots and many black & white images

TABLE 1
List of Features

Category	Type of Feature	Name of the Feature
High-Level Features	Content	Saliency—Classification between iconic and spread out image [30]
		Scene Features: PRICOLBP [60]
		Visual Clutter [63]
		Color versus Black & White Image [72]
High-Level Features	People	Object Detectors [14]
		Graphics Image versus Natural Image [56]
		# Females in the image [76]
		# Males in the image [76]
Low-Level Features	Affect	Face Detection—No. of faces and their orientation [frontal/pose], Relative area of the face wrt image [77]
		Head Detection [47]
		Upperbody Detection [12]
		Color Channel Statistics
	Aesthetics	Use of Light—Average Pixel intensity of V channel
		HSV Statistics—Mean of S channel and standard deviation of S, V channels; <i>Hue angular dispersion</i> in IHLS
		Emotion Statistics: Pleasure, Arousal and Dominance
		Colorfulness based on Earth Movers Distance
	Texture	Color Names—Amount of Black, Blue, Brown, Green, Gray, Orange, Pink, Purple, Red, White, Yellow
		Morphology based features-generic image descriptors including polynomial decompositions, high contrast features, pixel statistics, and textures [57]
		Low Depth of Field—Amount of focus sharpness in the inner part of the image
		Rule of Thirds—Mean of S,V channels in the inner rectangle of the image
Low-Level Features	Aesthetics	Regions—Average size of regions (after mean shift segmentation)
		Image Parameters (Size)
		Edge Statistics—Total number of edge points extracted with Canny
		Level of Detail (Watershed Segmentation)
	Texture	Wavelet Textures—Level of spatial graininess measured with a three-level Daubechies wavelet transform on the HSV channels
		Tamura Features—Amount of Coarseness, Contrast, Directionality
		GLCM Statistics—Amount of Contrast, Correlation, Energy, Homogeneity for each HSV channel
		Image Entropy

High-level features are described in Section 3 and low-level features are taken from [11].

(which is shown to be a sign of focus, subtlety of tones and versatility [64]). While users with a high score on extraversion have images with a lot of people, users with a low score have the opposite preference-consisting of scenic backdrops, without faces or objects to focus upon (images in the middle of the same collage). Also, people with a high score on neuroticism are characterised by anxious and tense behavior. This is conveyed by the images they like (images to the right of the collage), most of which have a high level of clutter [63].

Based on these observations, we hypothesize that personality traits should ideally be influenced by 'high-level' semantic concepts in images, each dealing with multiple facets of human behavior [23], [24], [27]. Therefore, we propose to use features such as identifying presence of people and understanding their attributes (gender, count, profile versus front view of face etc.) and also by representing the content of images by categorizing them as black & white versus color, quantifying the amount of clutter and scene characteristics.



Fig. 2. Need for including semantic features. Left: high score on conscientiousness; center: low score on extraversion; right: high score on neuroticism.



Fig. 3. Sample of the images 'liked' by users with different personality profiles. Left: high score on openness; center: high score on extraversion; right side: high score on agreeableness.

3.2 Relationship between Images 'liked' and Personality Profile

Here, we describe how 'high-level' semantic concepts relate to 'likes' of users with different personality profiles. Fig. 3 shows images liked by three users. The left part of the collage show images 'liked' by a user with a high score on Openness. These images show a bias towards dance, music and artistic photography. Images liked by a user having a high score on Extraversion are shown in the middle. This trait reflects an energetic approach towards the social and material world. This means that the user is expected to have a high affinity towards friends and partners—the images liked by the user can be seen to convey, in this case, a liking towards people of the opposite gender. Images on the right are liked by a user with a high score on Agreeableness, which reflects traits such as altruism, tender-mindedness, trust and, in behaviors like consoling friends who are upset etc. The images liked by the user convey a similar inclination.

Each of the five personality traits convey specific innate tastes of users which lead to certain behavioral attributes. Features extracted from the data can be used to detect the above-described behavioral attributes (which are less subtle than personality traits themselves), thereby accurately predicting users' personality.

Therefore, we propose to use features which can represent the high-level semantics in the image (such as the presence of people in images and their attributes, along with other content-related characteristics) apart from aesthetics and other low-level features extracted from the images. These high-level semantics are expected to give a better representation of a user's likes and thereby his/her personality.

3.3 Description of Features

After motivating the use of high-level features such as people- and content-oriented characteristics for representing users' likes, we describe the features we propose to use in our work.

3.3.1 People

Presence of people is a significant characteristic in images which influences several users to like an image. Research shows that human brain is particularly sensitive to faces, due to the presence of specific neural pathways [28] and hence the presence/absence can influence users with different traits (say Extraversion) to like/dislike images.

- *Head and Upperbody Recognition.* Perceptions of an image can depend significantly on the presence or absence of people in it. And persons' visibility is often partial in images, typically showing only the upper body. To address this, an upper-body detector [12] was used. And for the cases where only the head was in the field of view, head detector was used [47].
- *Face and Pose Recognition.* Where people in an image look (towards/away from the camera) also influences users' liking for images. Face detection and pose estimation by [77] was used. Pose angle characterizing profile/frontal face along with relative area of the face was taken.
- *Gender Identification.* To ascertain the distribution of gender in images, we used IntraFace [76]. We took support of manual annotation to prevent any errors.

3.3.2 Content

Users with different personality traits can have varying perceptions of an image based on the content. For instance, those with anxious and tense behavior seem to have different tastes from those who have an easygoing outlook [40], [58]. Also, achieving satisfactory performance in predicting self-assessments depends on 'Availability' [75] which means that the traits must be perceivable to others. Therefore, to represent the high-level semantic concepts from contents in images (which are easier to perceive) based on the motivations presented in Sections 3.1 and 3.2, we extract the following features:

- *Scene Features.* Scene classification has been one of the classical problems which researchers have been trying to address. The performance of this task is based on the features that are used. PRICoLBP feature [60] due to its effectiveness in classifying scenes.
- *Graphics Rendered Images.* Attractive computer rendered images often capture the attention of people, and they are exclusively posted and discussed on Flickr [15]. Also, natural images are liked by many. Therefore, to distinguish this cue, we used the feature in [56], to classify images as computer graphics or natural images.
- *Image Saliency.* Attention of users can be determined based on image saliency. When images having a scenic backdrop are compared with images with certain objects in the focus, the scale of image can be understood by its saliency. When there is nothing in the

TABLE 2
Notations Used in Formulation

N_C	number of classes
d	dimension of vector
$\mathbf{x}_j$	data instance j
y_j	label of instance j
I	indicator function
x_j^k	k th duplication of instance x_j
y_j^k	label of x_j^k
$\mathbf{w}$	weight vectors for SVM
$\hat{\mathbf{w}}$	augmented weighted vectors for OR
θ	threshold vectors for SVM
ρ	SVM margin
ϵ_i^k	loss of x_i^k
$\mathbf{d}$	Feature selection indicator vectors
α	Lagrange variables
$\mathcal{C}$	selected features index set
$\mathcal{D}$	feasible set of feature selection indicator vectors

image to capture the users' attention, the saliency is usually spread out. This would be the case with highly focused images and those which have a scenic backdrop. But images with clear objects show sharp saliency spikes, which is the usual case with iconic images or those with clear objects. To distinguish between both, saliency was used [30] to compute the mean maximal intensity points in images relative to a threshold.

- *Color versus Black and White Image.* To check if an image is black and white or a color image, we used the color naming technique proposed in [44]. This is a binary feature where 1 represents a color image and 0 represents a black and white image.
- *Visual Clutter.* Busyness of an image can be described by its clutter. As the number of objects in natural scenes is difficult to measure, clutter was measured using feature congestion metric developed in [63].

4 FEATURE SELECTION BASED ORDINAL REGRESSION (FS-OR) APPROACH

We define the notations we use for our formulation in Table 2. We are given the answers-images matrix, where the rows of the matrix represent the 10 answers to the BFI-10 questions and columns represent the different images. We can treat the classification of each answer as an ordinal regression (OR) problem. The advantage of OR over the normal regression is that OR considers the order information in the regression. Take the answer to the first question for example, 1 is least and 5 is strongest. In this case the decision function $f(C_1)$ should satisfy $f(C_1) < f(C_2) < f(C_3) < f(C_4) < f(C_5)$, where C_i represents the i th scale of given answer.

4.1 Converting Ordinal Regression Problem into Binary Classification Problem

To handle the proposed OR problem, we adopt the extended binary classification model [41]. In this model, an instance $\mathbf{x}$ is categorised as the i th ordinal class when the predicted output $h(\mathbf{x}) = \mathbf{w}^\top \mathbf{x}$ is in the range of $\theta_{i-1} < h(\mathbf{x}) \leq \theta_i$, where the ordered thresholds satisfying

$\theta_1 < \theta_2 < \dots < \theta_{N_C-1}$. To learn these thresholds, we first transform the training data instances $\mathbf{x}_j \in \mathbb{R}^d$ and their corresponding class label $y_j \in \{C_1, C_2, \dots, C_{N_C}\}, j = 1, \dots, n$ as:

$$\begin{aligned} \mathbf{x}_j^k &= [\mathbf{x}_j^\top, \mathbf{e}_k^\top]^\top \\ y_j^k &= 1 - 2I[y_j \leq k] \end{aligned} \quad (1)$$

for $k = 1, 2, \dots, N_C - 1$,

where $\mathbf{e}_k \in \mathbb{R}^{N_C-1}$ represents a vector with the k th entry as one and remaining entries as zero. Each training instance $\mathbf{x}_j$ is duplicated $N_C - 1$ times and relabelled into $N_C - 1$ binary cases. Notice that an extended binary instance $\mathbf{x}_j$ has a dimension of $d + N_C - 1$. Therefore, with the duplicated training data, one can train a classifier with an augmented weight vector $\hat{\mathbf{w}} = [\mathbf{w}^\top, -\theta^\top]^\top \in \mathbb{R}^{d+N_C-1}$, where $\theta = (\theta_1, \theta_2, \dots, \theta_{N_C-1})$. This model renders the prediction in a straightforward way as follows:

$$f(\mathbf{x}^k) = \text{sign}((\hat{\mathbf{w}}^\top \mathbf{x}^k)) = \text{sign}(h(\mathbf{x}) - \theta_k),$$

where $h(\mathbf{x}) = \mathbf{w}^\top \mathbf{x}$. Using Eq. 2, the predictive class label of instance $\mathbf{x}$ is given as follows:

$$\sum_{k=1}^{N_C-1} I[f(\mathbf{x}^k) = 1] + 1, \quad (2)$$

where $I(\cdot)$ is indicator function. To obtain the label prediction for an unlabeled data instance from the OR classifier, one projects the data instance onto the weight vector, and assigns a label by comparing the confidence value with N_{C1} threshold values. From the definition of $\hat{w}$, we know that θ is absorbed into $\hat{w}$ such that the optimization of SVM can automatically optimize θ and w simultaneously. For more detailed optimization proof, we request the readers to refer to reference [66].

4.2 Selecting Relevant Features

After extracting the high-level features, only a subset of features are related to each question. Therefore, we need to select related features which benefits most for OR in each question. We modify the Feature Generation Machine (FGM) [42], [69] to select the relevant features. Note that in Section 4, we have converted the OR problem into a binary classification problem. So we can directly apply FGM here with an additional constraint of indicator vector $\mathbf{d}$ specific for this application,

$$\begin{aligned} \min_{\mathbf{d} \in \mathcal{D}} \min_{\mathbf{w}, \epsilon, \rho} \quad & \frac{1}{2} \|\hat{\mathbf{w}}\|_2^2 + \frac{\lambda}{2} \sum_k^{N_C} \sum_{i=1}^n \epsilon_i^{k2} - \rho \\ \text{s.t.} \quad & y_i \hat{\mathbf{w}}^\top (\hat{\mathbf{x}}_i^k \odot \mathbf{d}) \geq \rho - \epsilon_i^k, \\ & i = 1, \dots, n; \quad k = 1, \dots, N_C - 1, \end{aligned} \quad (3)$$

where ρ denotes SVM margin, ϵ_i^k denotes loss of data $\mathbf{x}_i^k$, $\mathcal{D} = \{\mathbf{d} | \sum_{j=1}^d \mathbf{d}_j \leq B + N_C - 1, \mathbf{d}(d - N_{C+1} : \text{end}) = 1, \mathbf{d}_j \in \{0, 1\}\}$, $\mathbf{d}$ is the feature selection vector with entries 0, or 1 and B is parameter to control the number of selected features and $\mathbf{d}(d - N_{C+1} : \text{end}) = 1$ denotes last N_C element in $\mathbf{d}$ must be selected for thresholds for OR. In the following, we denote the proposed model feature selection based ordinal regression in short for FS-OR.

TABLE 3
Krippendorff's α Measuring Reliability
of Others-Assessment, Taken from [11]

Trait	O	C	E	A	N
α	0.07	0.16	0.23	0.19	0.20

4.3 Algorithm for Feature Selection Based Ordinal Regression

Observe that the above problem is an NP-hard problem. However, following the derivation of [69], we can convert the original problem into the following dual problem:

$$\max_{\alpha \in \mathcal{A}} \min_{\mathbf{d} \in \mathcal{D}} -\frac{1}{2} \left\| \sum_{i=1}^{(N_C-1)n} \alpha_i y_i (\mathbf{x}_i \circ \mathbf{d}) \right\|^2 - \frac{1}{2\lambda} \alpha' \alpha, \quad (4)$$

where $\mathcal{A} = \{\alpha \mid \sum \alpha_i = 1, \alpha_i \geq 0, i = 1, \dots, n\}$.

Further details on the optimization can be seen in Appendix A. Following the derivation of [69], this problem can be reduced into a Multiple Kernel Learning (MKL) problem, which can be solved by simple MKL [61] or its corresponding primal problem directly [68]. However, we observe that problem itself is very expensive due to the huge number of base kernels (see appendix). Fortunately, only small subset of $\mathcal{D}$ is active at optimality. With this the problem can be solved by the cutting plain algorithm [31], which can be divided into iterative two steps: 1) fixing $\mathcal{D}$ and updating α with MKL 2) fixing α and finding the active subset of $\mathcal{D}$.

Algorithm 1. Feature Selection Based Ordinal Regression (FS-OR).

Input: Training data $\{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_n, y_n)\}$, where $y_i \in \{C_1, C_2, \dots, C_{N_C}\}$.

Data Conversion: Converting Ordinal Regression Problem into binary class according to formulation.

1: Initialize α_i as $\frac{1}{n}$; Find the most violated features $\mathbf{d}$ and let $\mathcal{C} = \{\mathbf{d}\}$.

repeat

2: Initialize $\mu = 1$;

3: Update α by MKL.

4: Find the most violated features $\mathbf{d}$ and let $\mathcal{C} = \mathcal{C} \cup \{\mathbf{d}\}$

until Converged

Output: Selected features set $\mathcal{C}$.

Extended OR Training: We shrink the features as $\mathbf{x}^k(\mathcal{C})$ and feed data into SVM to train the model.

Prediction: For any test data $\mathbf{x}$, we first select features according to $\mathcal{C}$, namely $\mathbf{x}(\mathcal{C})$. Then substitute it into the following form:

$$f(\mathbf{x}) = \{k : \theta_{k-1} < h(\mathbf{x}) \leq \theta_k, \text{ where } h(\mathbf{x}) = \mathbf{w}^\top \mathbf{x}(\mathcal{C})\}. \quad (5)$$

Now the problem is how to find the most violated $\mathbf{d}$ from $\mathcal{D}$ each time with fixed α . To find active subset of $\mathcal{D}$, we have to solve an equivalent problem as mentioned below:

$$\max_{\mathbf{d} \in \mathcal{D}} \frac{1}{2} \left\| \sum_{i=1}^{(N_C-1)n} \alpha_i y_i (\mathbf{x}_i \circ \mathbf{d}) \right\|^2 - \frac{1}{2\lambda} \alpha' \alpha = \frac{1}{2} \sum_{j=1}^n c_j^2 \mathbf{d}_j, \quad (6)$$

where $c_j = \sum_{i=1}^{(N_C-1)n} \alpha_i y_i \mathbf{x}_{ij}$, α is obtained by MKL. Different from [69], our feasible set $\mathcal{D}$ of $\mathbf{d}$ has an additional constraint

TABLE 4
Spearman Correlations (Along with p-Values) between
Self-Assessed and Others-Assessed Personality Scores

	O	C	E	A	N
ρ	0.171	0.112	0.049	0.088	0.02
p	0.003	0.054	0.393	0.131	0.730

$\mathbf{d}(d - N_C + 1 : \text{end}) = 1$. Fortunately, we can still update the $\mathbf{d}$ by sorting first $d - N_C$ entries with fixing the last $N_C - 1$ entries as 1.

After learning the selected features index set $\mathcal{C}$, we further shrink the features according to $\mathcal{C}$ and feed data into SVM to train the model. For any test data $\mathbf{x}$, we first select features according to $\mathcal{C}$, namely $\mathbf{x}(\mathcal{C})$. Then substitute it into the following form:

$$f(\mathbf{x}) = \{k : \theta_{k-1} < h(\mathbf{x}) \leq \theta_k, \text{ where } h(\mathbf{x}) = \mathbf{w}^\top \mathbf{x}(\mathcal{C})\}. \quad (7)$$

The algorithm is depicted in Algorithm 1.

5 EXPERIMENTS

We describe results from experiments conducted to answer the research questions [RQ1, RQ2, RQ3] raised in the Introduction section of this paper. Section 5.2 deals with questions, RQ1 and RQ2 and Section 5.3 deals with question RQ3. Section 5.4 investigates the model predictive performance for a recommender application. To recapitulate, the research questions pertain to

- investigating the efficacy of proposed features set,
- evaluating the F2A+A2P personality modeling approach, and
- evaluating our contributions on a personality modeling based recommender application.

5.1 The PsychoFlickr Dataset

The PsychoFlickr dataset was collected by [11]. It consists of 60,000 images where 200 images each are liked by 300 'pro' Flickr users. For every user, personality assessment was done in two ways: (a) Self-assessed scoring, where users were asked to fill BFI-10 [62] questionnaire, recording their personality in terms FFM model traits. The users responded to items like "I am a reserved person" etc., (b) Others-assessed (attributed) scoring, where eight assessors were assigned to look at users' liked images and answer the BFI-10 questionnaire. The assessors responded to items like "This person is reserved", where the "person" is the user who 'liked' the images under examination. Thus every user has a collection of 200 'liked' images and also two sets of personality scores: one by the assessors and another by the users themselves. Apart from the dataset given by the authors of [11], we additionally downloaded the user profile information for all the users using the Flickr API.

The reliability of 'others-assessed' scores was measured using Krippendorff's α [38] (shown in Table 3). α is a robust to small sample sizes and the values reported in Table 3 are statistically significant and in line with those in literature [4], [34], [35]. Spearman correlations between corresponding traits of self-assessed and others-assessed personality scores are shown in Table 4. Self-assessments seem to reflect

TABLE 5

Automatic Personality Recognition Performance Comparison between Our Approach and [11] in Terms of Spearman Correlation Coefficient between Actual and Predicted Trait

Trait	Type	[11]	Our Approach
O	Self	0.25	0.39
	Attributed	0.35	0.49
C	Self	0.24	0.37
	Attributed	0.57	0.70
E	Self	0.28	0.43
	Attributed	0.62	0.77
A	Self	0.20	0.36
	Attributed	0.52	0.73
N	Self	0.14	0.29
	Attributed	0.60	0.75

All correlations for our approach are found to be significant ($p < 0.05$).

the actual personality of a user [62]. Assessment by others are a consequence of attributions and thus when provided by persons who are acquainted with users (like family members etc.) tend to have high correlations [7]. However, in this dataset, the assessment is done by total strangers (to verify the effect of Brunswick Lens model [8]), and thus the low correlations on some of the traits is expected. Taking this into consideration, we focus on the prediction of self-assessed personality traits. However, we report the results on others-assessed personality traits using our method in comparison to the current state-of-the-art. The latter results on others-assessed traits warrant further validation due to the nature of the dataset-low reliability scores (α).

Experimental Set Up. For all the experiments, unless specified otherwise, the data split is 75 percent training with 5-fold cross validation and 25 percent testing. Results are reported on test data. In the experiments, 'low-level features' refers to the features used in [11] and 'combined features' refers to the combined set of low-level features and high-level features (Table 1). This notation is used just for the sake of better discrimination of results between current state-of-the-art and our approach and not to define them as such. The traits, for which performance of models is being reported, are: conscientiousness (C), openness to experience (O), agreeableness (A), extroversion (E) and neuroticism (N).

5.2 Evaluating Modeling Approaches and Features

F2P Approach. LASSO regression was taken as the baseline (to compare with [11]). Second, to account for sparsity constraint (i.e., only a few features contributing to each answer as explained in Section 2), a sparse SVM modeling approach was chosen.

F2A+A2P Approach. Recent results [67] have shown that F2A+A2P gives better performance for modeling personality profiles of individuals inside video segments. We use our proposed FS-OR method in the context of F2A+A2P. FS-OR can also be used in the context of F2P and we evaluate its performance in the experiments below. In a previous version of our work [22], we used Sparse and Low-rank Transformation proposed by [67] in the context of F2A+A2P.

We trained F2P (Lasso and sparse SVR) and F2A+A2P (FS-OR) models using the low-level features as the input and

TABLE 6

Results of Predicting Self-Assessed Personality-Profiles for Both F2A+A2P (FS-OR) and F2P (LASSO and Sparse SVR) Approaches, Represented in Terms of (RMSE, Correlation)

With low-level features					
Trait	F2P			F2A+A2P	
	LASSO	sparse SVR	FS-OR	SLoT	FS-OR
O	1.69, 0.20*	1.77, 0.22*	1.76, 0.20*	0.98, 0.26*	0.97, 0.29*
C	1.78, 0.18	1.68, 0.16	1.67, 0.19*	0.90, 0.24*	0.89, 0.28*
E	2.07, 0.19*	1.83, 0.21*	1.82, 0.24*	0.97, 0.27*	0.97, 0.33*
A	1.66, 0.17	1.39, 0.18	1.27, 0.18*	0.83, 0.30*	0.83, 0.30*
N	2.20, 0.21	2.35, 0.13	2.38, 0.21*	1.20, 0.21*	1.19, 0.24*

With combined features					
Trait	F2P			F2A+A2P	
	LASSO	sparse SVR	FS-OR	SLoT	FS-OR
O	1.79, 0.24*	1.71, 0.23*	1.70, 0.26*	0.91, 0.32*	0.93, 0.39*
C	1.70, 0.23	1.65, 0.20*	1.63, 0.23*	0.83, 0.33*	0.89, 0.37*
E	1.85, 0.26*	1.82, 0.26*	1.81, 0.27*	0.90, 0.32*	0.92, 0.43*
A	1.44, 0.22*	1.37, 0.24*	1.28, 0.26*	0.72, 0.34*	0.78, 0.36*
N	2.29, 0.25	2.31, 0.20	2.32, 0.22*	1.11, 0.25*	1.17, 0.29*

Significant correlations ($p < 0.05$) are marked with *.

self-assessed personality scores as the output. Table 5 shows the performance (in terms of Spearman Correlation Coefficient) of our approach (FS-OR in the framework of F2A+A2P) when compared to the current state-of-the-art [11].

Table 6 shows that the F2A+A2P outperforms F2P approach with significant performance gains. Even the RMSE of traits using sparse SVR is lesser than that using LASSO. It also shows that F2A+A2P can bring huge gains in modeling personality scores, reducing the error by 40-50 percent. The same experiment was repeated using combined features. In Table 6, it is observed that in case of F2P approach, the RMSE for E using lasso with low-level features was 2.077 and the same using new features is 1.859. The improvement is about 11 percent. Similarly for other traits, adding new features led to a 3-12 percent increase in performance. There are cases where the performance has degraded as well (trait N, for instance), but the degradation was very small. We performed PCA on the set of features but the performance was less than the proposed approach—the correlation and RMSE between predicted and actual traits was lower by about 0.10-0.15 and 0.04-0.1 respectively for personality perception (predicting others-assessed scores) and by 0.30-0.50 and 0.1-0.2 respectively for personality recognition (predicting self-assessed scores). Here, it is seen that SLoT outperforms FS-OR in terms of RMSE while using combined features. However, in terms of correlation and accuracy (as seen in Table 7, FS-OR is seen to outperform SLoT).

Since the personality profiles are histograms over quantized bins it needs to be understood that the actual value of prediction error (deviation) can have a significant bearing on the above claim. More specifically, if the error is less than the quantization interval we can expect no error. The scores were divided into high, neutral and medium levels. Table 7 shows that F2A+A2P improves the performance by about 7-15 percent of predicting all the traits except for Openness. FS-OR is seen to outperform SLoT [67] in predicting all traits except Openness, where it is comparable. It

TABLE 7
Measuring Actual Deviations in Prediction of Personality Scores (Which Are Converted into High, Medium, Low Categories Based on Thresholding)

Trait	Classification Accuracy (in %)			
	F2P		F2A+A2P	
	sparse SVM	FS-OR	SLoT	FS-OR
O	62.4	62.7	66.10	63.1
C	58.1	70.9	70.50	79.5
E	53.7	72.4	69.70	84.7
A	64.3	74.5	72.30	87.5
N	50.7	61.2	61.50	65.5

should be noted that the images were ‘liked’ by Flickr ‘pro’ users, who pay a fee for using privileged features on Flickr. This means that the users’ interests would be highly inclined towards art, photography and the like, as mentioned in Section 3, thereby skewing the traits’ distribution causing low performance on Openness.

F2P approaches used in previous personality modeling literature show good performance for Extraversion and Conscientiousness traits as they can be easily perceived [29]. But when the approach of F2A+A2P, it results in significant gains in performance for all traits, as seen in the experiments. This can be attributed to the fact that the relationship between features and answers are less abstract. The performance gains can also be attributed to the usage of BFI scheme, a standard method used in psychology for personality assessment.

F2A+A2P can also be used to predict the users’ self-assessed scores, even without explicitly asking the users to answer the BFI-10 questionnaire. This can be a huge step forward in non-intrusive personality acquisition techniques, which has been shown to be preferred the most by users in [26].

5.3 Features Important for Self-Assessment

Table 8 reports the contribution of combined features that were learnt corresponding to each personality trait (based on the questions used to calculate the score on the corresponding trait) with the combined feature set and F2A+A2P modeling approach. These represent the weights allocated to the different features by our FS-OR model. For features which are multidimensional (vectors), this value represents the average of the weight on each dimension of the feature vector. Features omitted for different traits obtained a value of 0 in the trained models.

Users with a high score on the openness trait prefer colorful images [5] but do not like to have clutter or profile faces. Those on conscientiousness prefer black and white images with people—who are not facing towards them. Those with a high score on extraversion prefer images with a lot of people in them [43], [53], who are of the opposite gender (the negative weight on males is probably due to the higher number of males in the dataset). Users with high score on Agreeableness prefer spread out images with faces which are not profile. Neuroticism high score users prefer images with high clutter [48], some amount of graphics but not in color. The cited works also report similar findings by psychology researchers in other domains, verifying our results in the multimedia domain.

TABLE 8
Weights Allocated to Features by the Trained Model, Indicating Their Influence on Personality Traits Along with Pearson Correlation ($p < 0.05$) in *Italics* of the Feature Values with the Self-Assessed Traits

Trait	Positive	Negative
O	Scene(1.6, 0.08), Color(1.34, 0.07), #Upperbody(1.25, 0.05)	Clutter(−1.49, 0.03), #Profilefaces(−0.24, 0.01)
C	#Heads(1.41 0.04)	Color(−1.02 0.05), #Faces(−0.25, 0.01)
E	#Upperbody(2.79, 0.09), Female(1.45, 0.07), Saliency(1.25, 0.06)	Male(−0.38, 0.06)
A	Saliency(2.21, 0.10), #Faces(0.8, 0.04), Scene(0.63, 0.04)	#Profilefaces(−0.43, 0.03), Female(−0.13, 0.03)
N	Clutter(2.06, 0.11), Graphics(0.35, 0.04)	Color (−0.28, 0.03), Saliency(−0.329, 0.04)

These insights can help recommender systems to make quick and relevant decisions, in a cold-start scenario, as to what kind of images should be shown to a new user, whose personality is known. For example neurotic people were seen to ‘like images with clutter and graphics. To a new user, the system can show images whose scores on these two features are high. Then as the user keeps using the system, it can fine-tune the other aspects such as the user liking cluttered images with people or without people etc., which are more subjective in nature.

5.4 Recommender Application

The recommendation problem has two aspects:

- 1) Given a user, predict which image will the user like.
- 2) Given an image, predict which user (or group of users) will like it.

For the first aspect, our goal is, given a user profile, evaluate recall at top-K for all test images. Note that in the given dataset, we only have information about images that every user liked. We do not know if a user has seen the images liked by remaining users. Therefore, although we cannot reject the possibility that the images liked by other users could be his to his liking as well, we prefer to be more strict in our evaluation and the performance evaluation forms a lower bound on accuracy.

For these experiments, we used the models for both F2P (sparse SVM) and F2A+A2P as described in previous sections. We took a total of 2,000 images (for which user assignments are known and are different from the training set) as the test set and ranked the images based on RMSE between the predicted profile and users profile. The models were trained using 5-fold cross-validation setting on the training set of images and then used on the test set, aforementioned. Performance evaluation is based on recall for top-K (where $K \in \{1, 2, \dots\}$).

Fig. 5 shows that F2A+A2P is able to retrieve a higher number of images (in top-K) when compared with the F2P model. It is definitely a possibility that some images that were highly ranked by the model but were not counted towards recall might have been ‘liked’ by the user. To verify

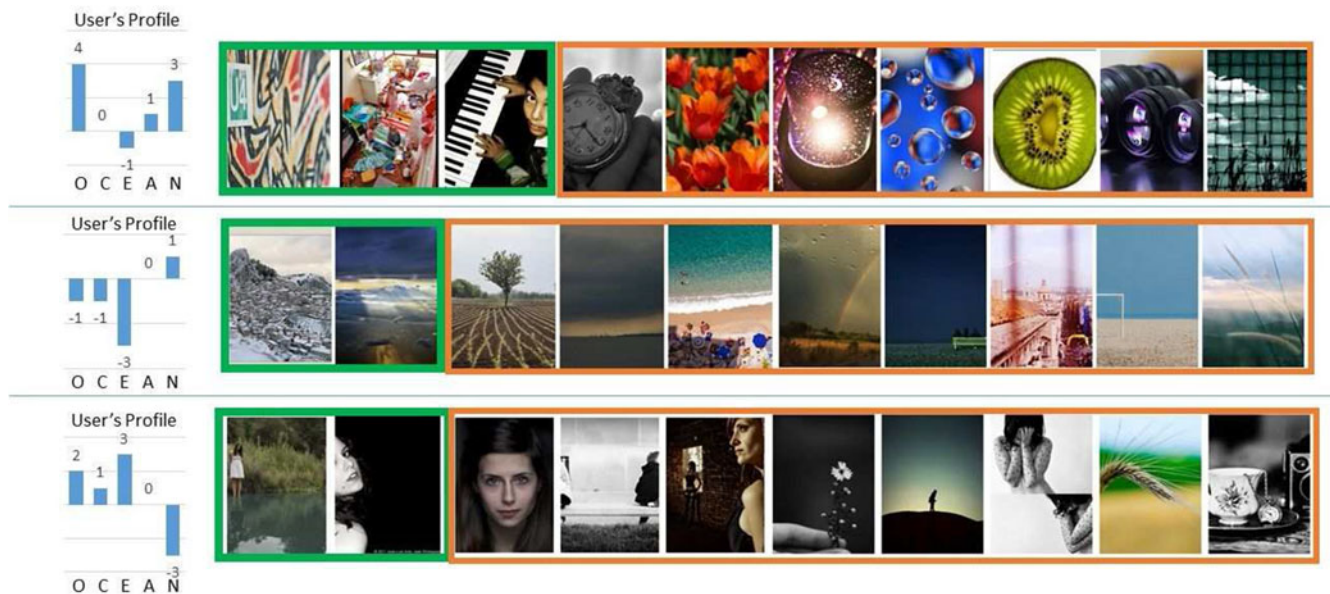


Fig. 4. Top 10 images retrieved by F2A+A2P for a sample of users based on their personality profile. The images with green border are 'liked' by the user. Retrieved images not belonging to the user's liked set can be seen to have lot of similarity w.r.t cues associated with the corresponding personality traits. Explained in Section 5.4.

this we visually inspected the top 10 images retrieved by the model for a few users, out of which three are shown along with the corresponding user profile in Fig. 4. The images bounded by green are 'liked' by the users. The first user has a high score on Openness and Neuroticism traits and the images are very colorful (which was found to be a common characteristic of images depicting custom work/home environments (image 2 from the left) and artistic photography). Some of the images retrieved have high clutter (images 1 and 10 from the left), which is a characteristic of images 'liked' by neurotic users. The second user has a very low score on extraversion and corresponding images retrieved have scenic backdrops without any explicit objects.

We also examined the difference in performance, given a set of images, of predicting the right user before and after adding our proposed features. The results are shown in Table 9—with combined features, performance of 25/75 percent data split is better than 50/50 percent split of low-level features.

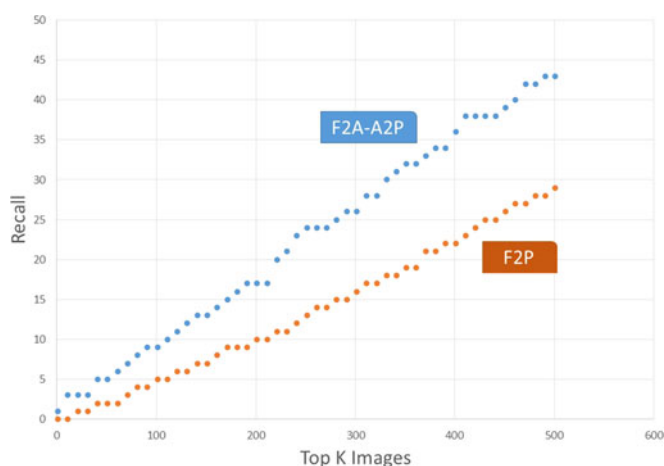


Fig. 5. Mean recall (for all users) using top k images retrieved by F2P and F2A+A2P models.

This confirms that a) representing images through semantic features improves performance in mapping images to users and b) the F2A+A2P approach is highly effective in modeling personality profiles from images with high generalization.

For the second aspect of image recommendation, our goal is, given a set of users and an image, to study what kind of user (or group of users) would like an image. This involves, first predicting the profile from the image and then comparing with all user profiles to compute the RMSE. The image gets assigned to the user with the lowest RMSE. Then we can check if the image is correctly assigned. This is done for every user and we find out the accuracy. Note that images can portray multiple traits and hence it is more useful to target user groups rather than individual users.

To test for user group assignments we use the Flickr API to extract the profile information (gender, ethnicity) of the users using their User IDs/User names and grouped them accordingly. When the data was collected by [11], 75 percent of the users were males and 25 percent were female. 75 percent of the users were European, 15 percent were Americans and 10 percent were Asians. The performance (confusion matrix) of the model in predicting the gender of the user who would like the images is shown in Table 11 and the

TABLE 9
Given a Set of Images, Performance of Using Combined Features at Predicting the Right User

Train/Test split:		low-level features		
		75%/25%	50%/50%	25%/75%
Correct user hit	top 5	75.60%	70.9	69.3
	top 3	61.10%	57.2	54
Train/Test split:		combined features		
		75%/25%	50%/50%	25%/75%
Correct user hit	top 5	78.30%	71.9	73
	top 3	65.20%	58.2	58.8

TABLE 10
Confusion Matrix for Mapping Images
to Users' Ethnic Profiles

	Europeans	Americans	Asians
Europeans	93.13	6.87	0.00
Americans	16.67	83.33	0.5
Asians	0.00	18.00	82.00

same for ethnicity is shown in Table 10. We see that there is a considerable misclassification between American and European ethnicities; however, none of the them were misclassified as Asian, possibly indicating the similarities between American and European cultures and the distance from Asian cultural preferences [10], [13], [49].

As an alternate way of grouping, we clustered the users into five groups based on the users' personality profiles (by taking the correlation between the scores of 5-personality scores as the distance metric). Clusters 1 through 5 consisted of 5, 30, 15, 30, and 20 percent users respectively. We performed the same experiments (as described above with gender and ethnicity) to check if the images go to the right cluster of users and checked the assignment accuracy. The results are reported in Table 12. Certainly the personality clusters could be formed in a better manner than we did, but this is a first attempt to understand if personality modeling based image recommendation is a promising approach.

6 CONCLUSIONS

In this paper, the problem of computationally predicting users' self-assessed personality profiles along the Big-5 traits, based on images they 'like' was studied.

To answer the RQ1 raised at the outset of this paper, the results of personality recognition improved considerably from the current state-of-the-art by using combined features. This can be attributed to the observation that high Availability criteria [75], which indicates that traits perceivable by others help improve the personality recognition performance. This was possibly a consequence of using features focused on the content and people in the image. This gives an indication that predicting users' personality, which is a challenging task, we need more human-interpretable and semantic representations.

With regards to RQ2, it was seen that the approach of predicting answers to the questions to personality inventory (F2A+A2P), BFI-10 in this case, and then converting them to personality scores outperforms the approach of predicting personality scores directly from the image features (F2P). This can be used to facilitate non-intrusive personality acquisition techniques, which have been shown to be highly preferable [26].

With regards to RQ3, it was seen that features related to scene, color, #uppebody have a positive influence and

TABLE 11
Confusion Matrix for Mapping Images to Gender

	Male	Female
Male	95.2	4.80
Female	10.00	90.00

TABLE 12
Confusion Matrix for Mapping Images
to Personality Profile Clusters

	C1	C2	C3	C4	C5
C1	7.00	0.00	0.00	93.00	0.00
C2	0.00	50.33	0.5	26.67	22.5
C3	0.00	0.00	76.67	22.00	1.33
C4	0.00	0.16	21.33	78.16	0.33
C5	0.00	0.00	0.00	49.5	50.5

clutter, #profilefaces have a negative influence in predicting the openness trait. #Heads contributes positively and color, #faces contribute negatively to conscientiousness. #Upperbody, female, saliency contribute positively and male contributes negatively to extraversion. Saliency, #faces, scene have a positive influence and #profilefaces, female have negative influence on agreeableness. Clutter, graphics contribute positively and color, salience negatively to neuroticism. These insights could be used in the application of image recommendation to use personality as a good way of providing quick 'likable' recommendations to users. This could address the cold-start situation where information about users is missing. However, in this scenario, further work is warranted with datasets that have both 'like' and 'dislike' annotations so as to have a more concrete evaluation.

Further, in this work all the features focused on the visual content and they were seen to model users' preferences quite well. However, the tags associated with images might complement visual features by conveying information which visual features might fail to convey [33]. The place in the image or the relationship with the person in the image, or the scene in the image might trigger user's memory, and this context is often present in tags [18]. It might be worthwhile to explore the contribution of tags to personality prediction. Inspired by the [79], [80] we may explore the possibility of transfer knowledge from other modalities (e.g., text) into domain of interest (e.g. image) to further improve the performance.

Attributed traits are often predictive of important aspects of social life, and can be useful in multiple domains [36], [37]. Therefore, another aspect of future work would involve collecting more 'reliable' survey data for personality profiles using larger and possibly multiple inventories, especially with regards to others-assessed personality profiles due to low reliability scores (Table 3). This makes it difficult to compare the prediction performance between the self-assessed and others-assessed profiles. Additionally, research has shown that emotion plays a crucial role in users' 'likes' [1], [73]—exploring this as a latent feature for understanding the relationship between personality and 'likes' might give deeper insights into 'who likes what and, why'.

APPENDIX A OPTIMIZATION FORMULATION FOR FS-OR

By bringing an additional variable τ , the above optimization further becomes:

$$\max_{\alpha \in \mathcal{A}, \tau} -\tau : \tau \geq -L(\alpha, \mathbf{d}^t), \forall \mathbf{d}^t \in \mathcal{D}, \quad (8)$$

where is a convex Quadratically Constrained Quadratic Programming (QCQP) problem. Define $\mu_t \geq 0$ as dual variable for each constraint. Its Lagrangian can be rewritten as:

$$\max_{\alpha \in \mathcal{A}} \min_{\mu \in \mathcal{M}} \sum_{\mathbf{d}^t \in \mathcal{D}} \mu_t S(\alpha, \mathbf{d}^t) \quad (9)$$

$$= \min_{\mu \in \mathcal{M}} \max_{\alpha \in \mathcal{A}} (\alpha \odot \mathbf{y})^\top \left(\sum_{\mathbf{d}^t \in \mathcal{D}} \mu_t \mathbf{X}_t \mathbf{X}_t^\top + \frac{1}{C} \mathbf{I} \right) (\alpha \odot \mathbf{y}), \quad (10)$$

where $\mathbf{X}_t = [\mathbf{x}_1 \odot \mathbf{d}^t, \dots, \mathbf{x}_{(N_C-1)n} \odot \mathbf{d}^t]$ and $\mathbf{I}$ denotes identity matrix.

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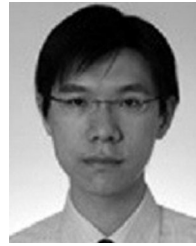
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